

Underwater Vision-Based Surveillance using AI and ML

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Day 7 — Initial Dataset Inspection & Preliminary Dataset Report

1. Objective

To inspect the three supplied underwater datasets, verify their structure and annotations, identify missing or inconsistent data, examine duplicates and class coverage, and preserve an untouched raw copy before further preprocessing.

2. Datasets Inspected

Dataset	Classes	Train	Validation	Test
DIATAquarium	11	8121	773	Not present locally
Aquatic Plant	1	1892	179	89
Well	4	3795	191	184

All three datasets were inspected separately. The local DIATAquarium dataset contains train and validation folders, but no separate test folder.

3. Raw-Data Archive

An untouched backup of the supplied dataset was created.

Backup location: Google Drive/Dataset_RAW_BACKUP

The raw backup will not be modified. Future preprocessing will be performed on a separate working copy.

4. Dataset Structure

Each dataset contains:

- images/ — image files
- labels/ — YOLO annotation files
- data.yaml — dataset configuration
- train/, valid/, and test/ folders where available

5. Image and Label Inspection

Dataset	Split	Images	Labels
DIATAquarium	Train	8121	8121
DIATAquarium	Valid	773	773
Aquatic Plant	Train	1892	1879
Aquatic Plant	Valid	179	179
Aquatic Plant	Test	89	89
Well	Train	3795	3780
Well	Valid	191	191
Well	Test	184	184

6. Image-Label Pairing

- DIATAquarium: 0 unmatched images and 0 unmatched labels.
- Aquatic Plant: 13 training images without matching labels.
- Well: 15 training images without matching labels.
- No labels were found without corresponding images.

The unmatched images were visually inspected. They will not be automatically deleted because some may represent background or negative samples.

7. YOLO Annotation Validation

YOLO annotations were checked for format, numeric values, normalized bounding-box coordinates, empty labels and invalid entries.

Finding: No invalid YOLO annotation entries were detected. Empty labels may represent valid background or negative images.

8. Initial Visual Inspection

Sample images and annotations were visually inspected. Main observations:

- Haze and low visibility in some underwater images.
- Reduced contrast.
- Some images contain little or no clearly visible object.
- Sampled bounding boxes generally correspond to visible objects.

9. Class / Target Coverage

- DIATAquarium: 11 classes with uneven object distribution.
- Aquatic Plant: 1 class.
- Well: 4 classes with severe class imbalance; fishes is dominant, while Inlet-pipe and stone are very rare.

No balancing or deletion was performed during the initial inspection.

10. Duplicate Check

A hash-based first-look duplicate check was performed using samples.

Dataset	Split	Result
DIATAquarium	Train	0 duplicates / 100
DIATAquarium	Valid	0 duplicates / 100
Aquatic Plant	Train	6 duplicates / 100
Aquatic Plant	Valid	0 duplicates / 100
Aquatic Plant	Test	0 duplicates / 89
Well	Train	0 duplicates / 100
Well	Valid	0 duplicates / 100
Well	Test	0 duplicates / 100

The 6 duplicates found in the Aquatic Plant training sample were identified as a preliminary finding. A full duplicate/near-duplicate check is planned before final dataset freezing.

11. Metadata Observation

Several filenames contain patterns such as mp4 and frame numbers, indicating that some images may have originated from video-frame extraction. This should be considered when checking for possible source-video or frame leakage between dataset splits.

12. Initial Dataset Risks

Risk	Initial Finding
Missing labels	13 Aquatic Plant + 15 Well training images
Empty annotations	Present in all datasets
Invalid annotations	None detected
Class imbalance	Strong in Well; uneven in DIATAquarium
Low visibility	Haze, low contrast and reduced visibility
Different resolutions	Present in DIATAquarium — sampled train images: 1920×1080; sampled validation images: 2048×1536. Aquatic Plant and Well sampled images: 640×640.
Duplicate images	6 found in Aquatic Plant training sample
Source-video leakage	Possible due to video-frame filename patterns
Corrupt / unreadable images	No corrupted images were found in the sampled image-quality audit.
DIAT test split	No local test folder. Split handling will follow the mentor-approved strategy before final dataset preparation.

13. Preliminary Data-Preparation Plan

- Preserve the untouched raw dataset.
- Do not automatically delete images without labels.
- Review missing-label images before making corrections.
- Retain legitimate empty/background annotations.
- Standardize image size during preprocessing.
- Review duplicate and near-duplicate images before final split freezing.
- Check source-video/frame leakage before final split freezing.
- Consider class imbalance during model training and evaluation.
- Resolve DIATAquarium split handling according to the mentor-approved strategy before final dataset preparation.

14. Preliminary Findings

- Dataset structure, image counts and label counts were recorded.
- Image-label pairing was checked. And YOLO annotations were validated.
- Missing-label and empty-label cases were identified.
- Initial class imbalance and underwater image-quality issues were observed.
- A first-look duplicate check was completed.
- Metadata patterns suggesting video-frame extraction were observed.
- No irreversible data cleaning was performed.
- An untouched raw backup was created for reproducibility.